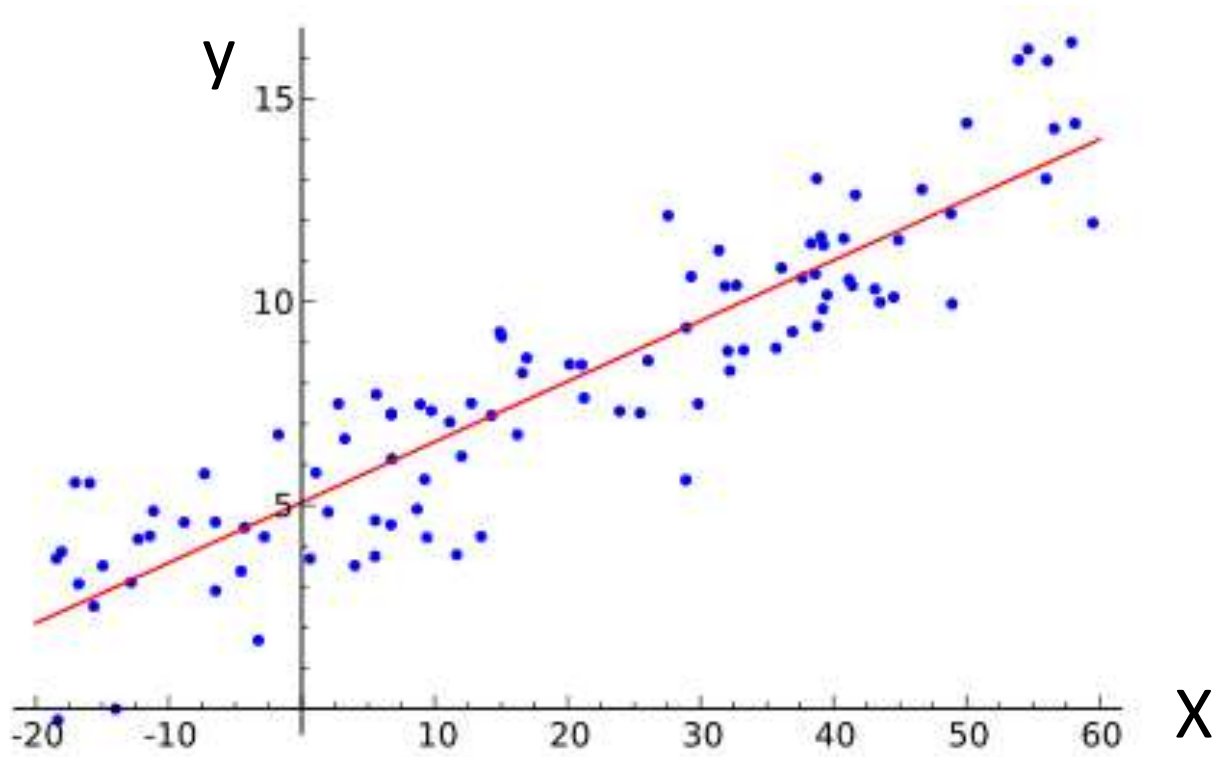


Linear Regression

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Linear Regression

We want to find the best line (linear function $y=f(X)$) to explain the data.



Example Data set for Regression

Sample ID	Engine Size	Cylinder	Fuel Consumption	CO2 Emission
1	2.0	4	8.5	196
2	2.4	4	9.6	221
3	1.5	4	5.9	136
4	3.5	6	11.1	255
5	3.5	6	10.6	244
6	3.5	6	10.0	230
7	3.5	6	10.1	232
8	3.7	6	11.1	255
9	3.7	6	11.6	269
10	2.4	4	9.2	??

Using Linear Regression to predict continous values

Linear Regression Topology

- Simple Linear Regression:
 - Predict **CO2Emission** according to **Engine Size** of all cars
 - Independent variable (x): Engine Size
 - Dependent variable (y): CO2Emission
- Multiple Linear Regression:
 - Predict **CO2Emission** according to **Engine Size** and **Cylinder** of all cars
 - Independent variable (x): Engine Size, Cylinder, etc.
 - Dependent variable (y): CO2Emission

Simple Linear Regression with one independent variable

Sample ID	Engine Size	Cylinder	Fuel Consumption	CO2 Emission
1	2.0	4	8.5	196
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3	1.5	4	5.9	136
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Input variable x

Output variable y

$$\hat{y}_1 = \beta_0 + \beta_1 x_1$$

Multiple Linear Regression

The predicted value of y is given by:

$$\hat{y} = \hat{\beta}_0 + \sum_{j=1}^p X_j \hat{\beta}_j$$

The vector of coefficients $\hat{\beta}$ is the regression model.

If $X_0 = 1$, the formula becomes a matrix product:

$$\hat{y} = X \hat{\beta}$$

Linear Regression

We can write all of the input samples in a single matrix **X**:

$$\text{i.e. rows of } \mathbf{X} = \begin{pmatrix} X_{11} & \cdots & X_{1n} \\ \vdots & \ddots & \vdots \\ X_{m1} & \cdots & X_{mn} \end{pmatrix}$$

are **distinct observations**, **columns of X** are **input features**.

Residual Sum-of-Squares

To determine the model parameters $\hat{\beta}$ from some data, we can write down the Residual Sum of Squares:

$$\text{RSS}(\beta) = \sum_{i=1}^N (y_i - \beta x_i)^2$$

or symbolically $\text{RSS}(\beta) = (\mathbf{y} - \mathbf{X}\beta)^T (\mathbf{y} - \mathbf{X}\beta)$. To minimize it, take the derivative wrt β which gives:

$$\mathbf{X}^T (\mathbf{y} - \mathbf{X}\beta) = 0$$

And if $\mathbf{X}^T \mathbf{X}$ is non-singular, the unique solution is:

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

Iterative Regression Solutions

The exact method requires us to invert a matrix $(\mathbf{X}^T \mathbf{X})$ whose size is nfeatures x nfeatures. This will often be **too big**.

There are many gradient-based methods which reduce the RSS error by taking the **derivative wrt** β

$$\text{RSS}(\beta) = \sum_{i=1}^N (y_i - \beta x_i)^2$$

which was

$$\nabla = \mathbf{X}^T (\mathbf{y} - \mathbf{X}\beta)$$

Stochastic Gradient

A very important set of iterative algorithms use **stochastic gradient** updates.

They use a **small subset or mini-batch** $\mathbf{X}$ of the data, and use it to compute a gradient which is added to the model

$$\beta' = \beta + \alpha \nabla$$

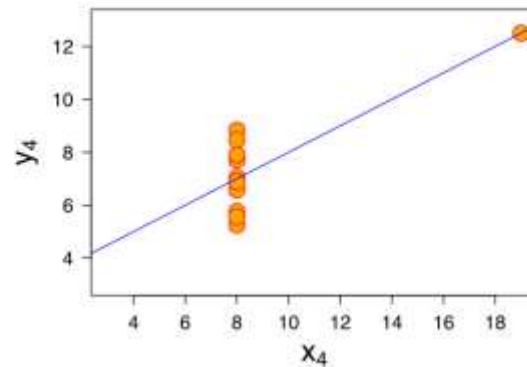
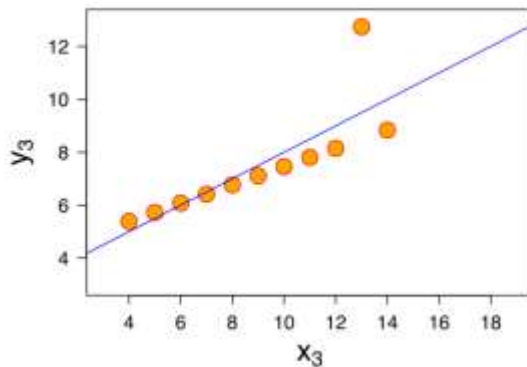
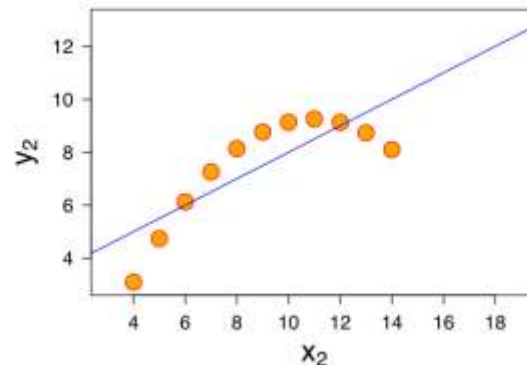
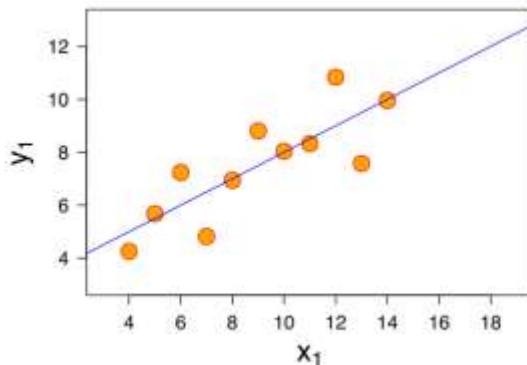
Where α is called the **learning rate**.

These updates happen **many times** in one pass over the dataset.

Its possible to compute high-quality models with very few passes, sometime with less than one pass over a large dataset.

R^2 -values and P-values

We can **always** fit a linear model to any dataset, but how do we know if there is a **real linear relationship**?



R²-values and P-values

Approach: Use a hypothesis test. The null hypothesis is that there is no linear relationship ($\beta = 0$).

Statistic: Some value which should be small under the null hypothesis, and large if the alternate hypothesis is true.

R-squared: a suitable statistic. Let $\hat{y} = X \hat{\beta}$ be a predicted value, and $\bar{y}$ be the sample mean. Then the R-squared statistic is

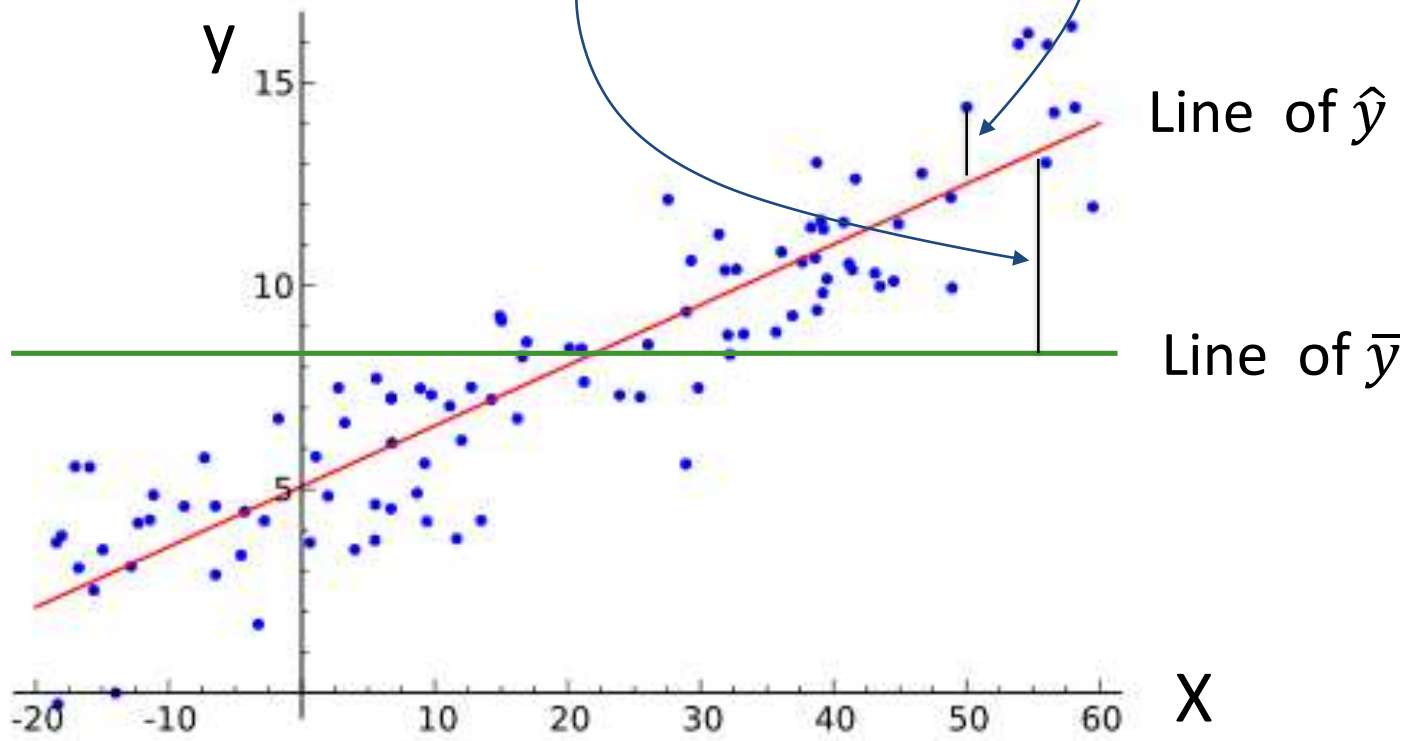
$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

And can be described as the fraction of the total variance not explained by the model.

R-squared

Small if good fit

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$



R^2 -values and P-values

Statistic: From R-squared we can derive another statistic (using degrees of freedom) that has a standard distribution called an **F-distribution**.

From the CDF for the F-distribution, we can derive a **P-value** for the data.

The P-value is, as usual, the probability of observing the data under the null hypothesis of no linear relationship.

If **p is small**, say less than 0.05, we conclude that **there is a linear relationship**.